

Energy, Heat and Work

Answer key with worked solutions

A. Ideas

1. Definitions and systems.

(a) **Energy**: the capacity to do work or transfer heat. **Heat**: energy transferred because of a temperature difference. **Work**: energy transferred by a force acting through a distance (e.g. a gas pushing a piston).

(b) Boiling pan: **open** (steam and heat escape). Sealed balloon: **closed** (heat can pass, gas cannot). Insulated sealed flask: **isolated** (ideally, neither passes).

2. Bath and cup.

The **cup** has the higher temperature (100 °C). The **bath** contains more thermal energy, because it has far more water: temperature reflects the **average** energy per particle, while the total thermal energy also depends on how many particles there are.

B. Kinetic and potential energy

3. Kinetic energy.

(a) $E_k = \frac{1}{2} \times 2.00 \text{ kg} \times (3.00 \text{ m/s})^2 = \mathbf{9.00 \text{ J}}$

(b)

$$\begin{aligned} E_k &= \frac{1}{2} \times 4.65 \times 10^{-26} \text{ kg} \\ &\quad \times (515 \text{ m/s})^2 \\ &= \mathbf{6.17 \times 10^{-21} \text{ J}} \end{aligned}$$

4. Potential energy.

$$E_p = mgh = 5.00 \text{ kg} \times 9.81 \text{ m/s}^2 \times 2.00 \text{ m} = \mathbf{98.1 \text{ J}}$$

C. Energy units

5. Conversions.

$$\begin{aligned}\text{(a) } 520 \text{ kcal} \times \frac{4.184 \text{ kJ}}{1 \text{ kcal}} &= 2175.68 \text{ kJ} = \mathbf{2.18 \times 10^3 \text{ kJ}} \\ \text{(b) } 1200 \text{ kJ} \times \frac{1 \text{ kcal}}{4.184 \text{ kJ}} &= \mathbf{287 \text{ kcal}} \\ \text{(c) } 2.5 \text{ kWh} \times \frac{3.6 \times 10^6 \text{ J}}{1 \text{ kWh}} &= \mathbf{9.0 \times 10^6 \text{ J}}\end{aligned}$$

D. Work and the first law

6. Compression.

$$\begin{aligned}\Delta V &= 4.0 \text{ L} - 10.0 \text{ L} = -6.0 \text{ L} \\ w &= -P_{\text{ext}} \Delta V = -2.00 \text{ atm} \times (-6.0 \text{ L}) \\ &= +12 \text{ L atm} \times \frac{101.325 \text{ J}}{1 \text{ L atm}} \\ &= \mathbf{+1.2 \times 10^3 \text{ J}}\end{aligned}$$

Positive: the surroundings do work **on** the gas, so the system gains energy.

7. ΔU .

$$\begin{aligned}\text{(a) } \Delta U &= (-300. \text{ J}) + (+150. \text{ J}) = \mathbf{-150. \text{ J}} \\ \text{(b) } \Delta U &= (+500. \text{ J}) + (-200. \text{ J}) = \mathbf{+300. \text{ J}} \\ \text{(c) Rigid container: } \Delta V &= 0, \text{ so } w = 0 \text{ and } \Delta U = q = \mathbf{+75 \text{ J}}\end{aligned}$$

E. Challenge

8. Piston.

$$(a) \Delta V = 6.00 \text{ L} - 1.00 \text{ L} = 5.00 \text{ L}$$

$$w = -1.00 \text{ atm} \times 5.00 \text{ L} = -5.00 \text{ L}\cdot\text{atm}$$

$$\times \frac{101.325 \text{ J}}{1 \text{ L}\cdot\text{atm}} = -507 \text{ J}$$

$$(b) q = +2.50 \text{ kJ} = +2500 \text{ J}$$

$$\Delta U = q + w = (+2500 \text{ J}) + (-507 \text{ J})$$

$$= +1993 \text{ J} = +1.99 \text{ kJ}$$

(c) ΔU is positive, so the internal energy **increased**: the gas gained more energy as heat than it lost as work.